Astronomy - Graduate Level Assessment 2 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which is not an essential requirement for life as we know it?
   (a) The ability to breathe oxygen
   (b) The ability to make energy from sunlight or to eat things that do
   (c) A source of organic molecules
   (d) A and B
2. Why do we look for water-ice in craters at Mercury's pole?
   (a) Actually water-ice is all over Mercury and not just at the poles.
   (b) The pole is the only place fortunate enough to have had comet impacts
   (c) Radar from the earth can only see Mercury's poles.
   (d) These craters contain the only permanently shadowed regions on Mercury
3. Which of these has NOT been one of the main hypotheses considered for the origin of the Moon?
   (a) The Moon split from the Earth due to tidal forces.
   (b) The Moon was captured into Earth orbit.
   (c) The Earth and Moon co-accreted in the solar nebula.
   (d) Earth was rotating so rapidly that the Moon split from it.
4. The distance between the Earth and the star Altair is one million times greater than the distance between the Earth and the Sun. How far is Altair from the Earth?
   (a) 9.3 x 1013 meters
   (b) 9.3 x 1010 meters
   (c) 1.5 x 1014 meters
   (d) 1.5 x 1017 meters
5. When was the telescope invented by Galileo?
   (a) 1409

(b) 1509
(c) 1609
(d) 1709

True / False (10 marks)

Answer True or False

6. True or False: The ecliptic refers to the axial tilt of the Earth, which affects the seasons throughout the year.
7. True or False: The number of sunspots on the Sun increases approximately every 11 years, a period known as the solar cycle.
8. True or False: The diameter of Pluto was first accurately measured using brightness measurements during mutual eclipses with its moon Charon.
9. True or False: Type Ia supernovae occur in binary systems, typically involving a white dwarf accreting material from a companion star.
10. True or False: A comet's tail always points away from the Sun due to the solar wind and radiation pressure acting on the comet's gas and dust.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the methodology employed by Eratosthenes to estimate the Earth's size in 240 B.C., including the significance of his observations in different latitudes.
12. Discuss the historical significance and scientific contributions of the U.S.S.R.'s lander missions to Venus, analyzing their impact on our understanding of the planet.

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